

A new limit on k -core integer partitions and the strong TASEP^(k)

Svante Linusson

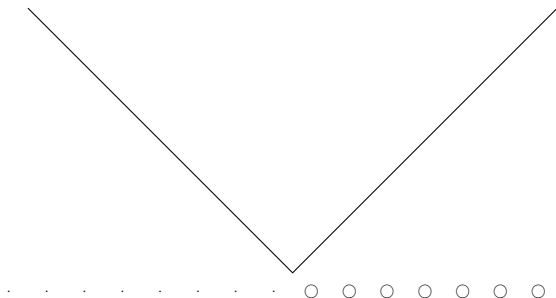
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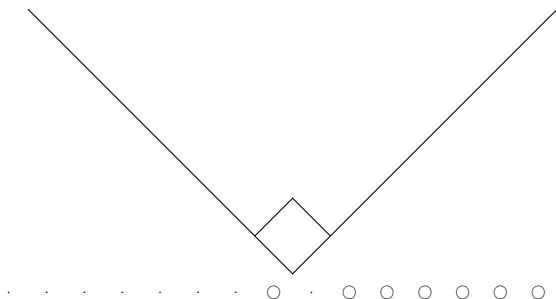
Jump process for growing a partition

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All positive positions are occupied with a particle and all non-positive are empty. Particles can jump left if that site next is empty. A particle jumping is the same as adding a new box in the partition.



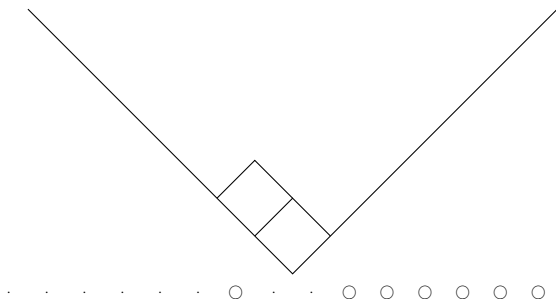
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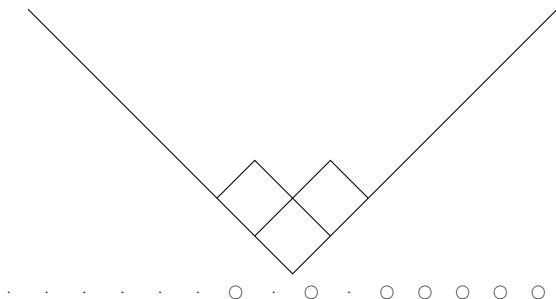
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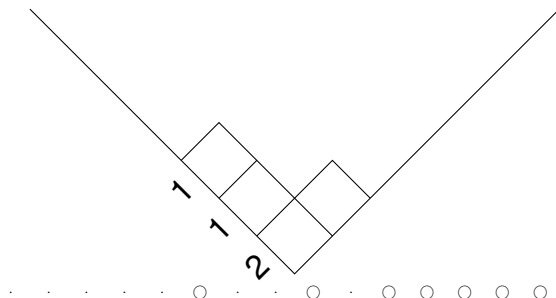
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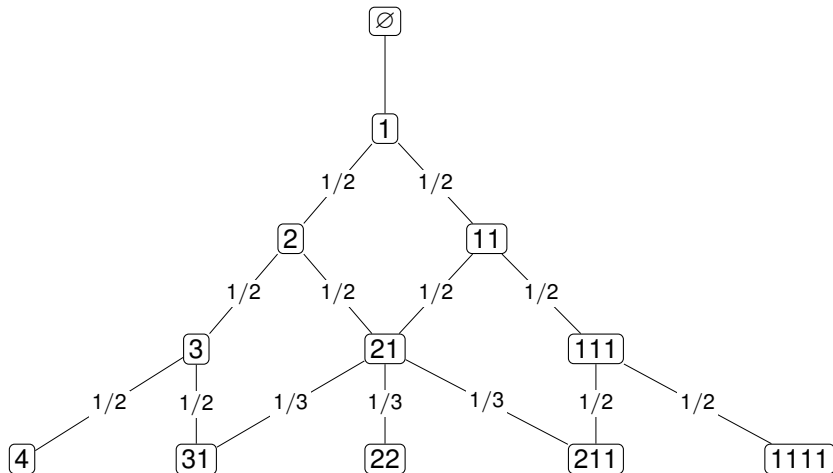
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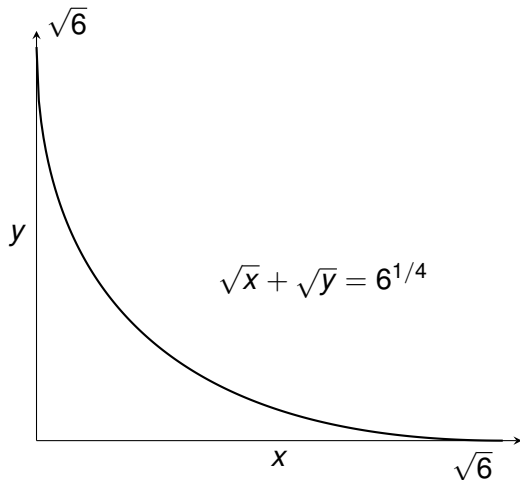
Random partitions, boxes added uniformly

One classical model is to add a new box uniformly at each step.



Random partitions, boxes added uniformly

The limit shape was studied by Rost '81.



Kurt Johansson '00 and others have famously studied its fluctuation.

Random partitions, plancherel measure

Let f_λ be the number of chains from $\emptyset$ to λ in the Young's lattice. Also the number of Standard Young Tableaux. E.g. $f_{3,1} = 3$ since

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1	3	4

3		
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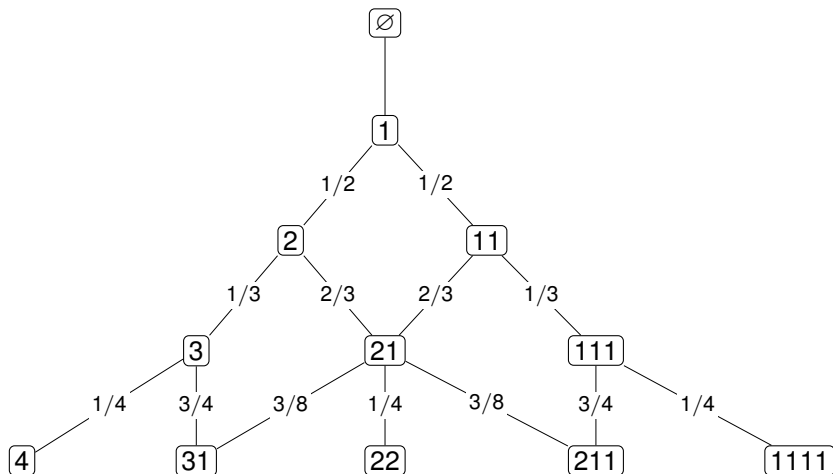
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Plancherel growth process has $P(\lambda \rightarrow \Lambda) = \frac{f_\Lambda}{nf_\lambda}$,

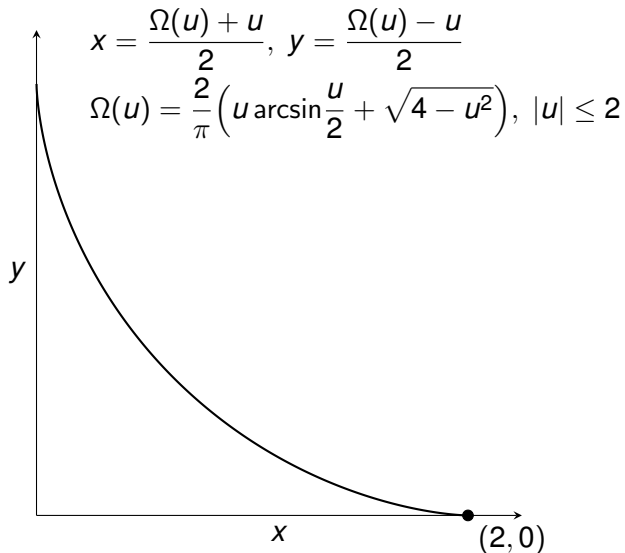
we get measure $\pi(\lambda) = \frac{f_\lambda^2}{n!}$, since

$$\pi(\Lambda) = \sum_{\lambda \rightarrow \Lambda} \pi(\lambda) P(\lambda \rightarrow \Lambda) = \sum_{\lambda \rightarrow \Lambda} \frac{f_\lambda^2}{(n-1)!} \frac{f_\Lambda}{nf_\lambda} = \frac{f_\Lambda^2}{n!}$$

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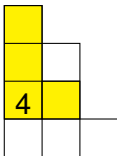
Random partitions, plancherel measure



The Logan-Shepp-Vershik-Kerov curve.

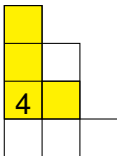
k -cores

In an integer partition, the **hook length** is the number of squares to the right or above a given square.



k -cores

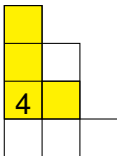
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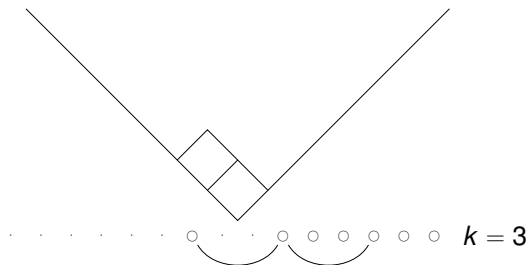
Content of a cell $c = (i, j)$ in the i th row and the j th column is $j - i$. For k -cores we consider the content modulo k . Here $k = 5$.

3

4	0											
0	1											
1	2	3										
2	3	4	0	1								
3	4	0	1	2	3							
4	0	1	2	3	4	0	1	2				
0	1	2	3	4	0	1	2	3	4	0	1	2

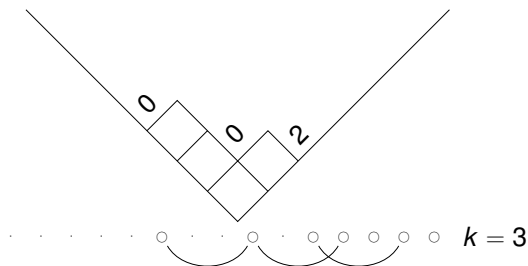
mod k jump process

Now all positions at position $i \pmod k$ try to jump at the same time, for some $0 \leq i < k$.



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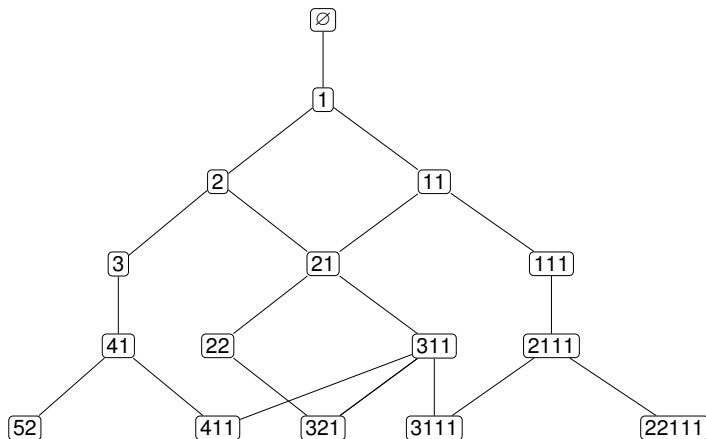
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This will grow a k -core.

k -cores

The weak order of 4-cores. Graded by number of times boxes has been added.



Let $f_{\kappa}^{(k)}$ (the weak dimension) be the number of paths to the k -core κ .

k -cores grown with uniform measure

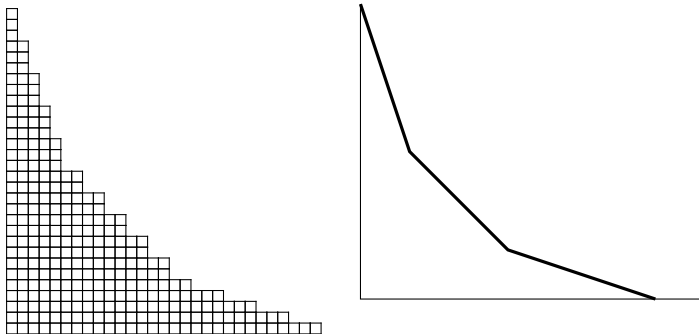
Let D_k be the piecewise linear curve with vertices $v_i = \gamma \left(\binom{i}{2}, \binom{k-i+1}{2} \right)$, for $i = 1 \dots k$, where $\gamma = \frac{2\sqrt{6}}{n\sqrt{n^2-1}}$ is a scaling constant.

Theorem (Ayyer-L. 2012, Conjectured by Lam)

The limit shape of the uniform growth of k -cores is D_k , scaled appropriately.

k -cores

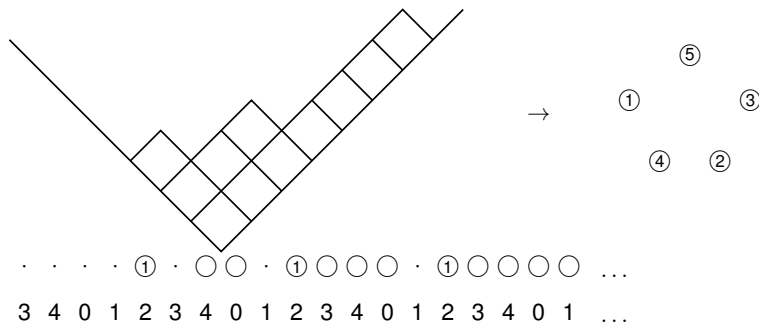
A large 4-core and the limiting shape.



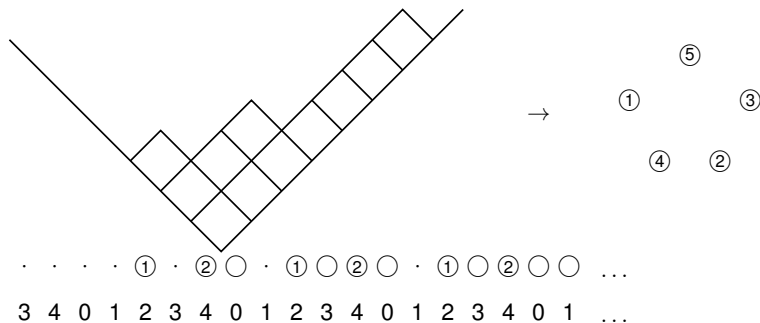
Note that the slopes are deterministic.

The vertices converges to be on the limit shape by Rost mentioned first. A finite version of Rost's curve.

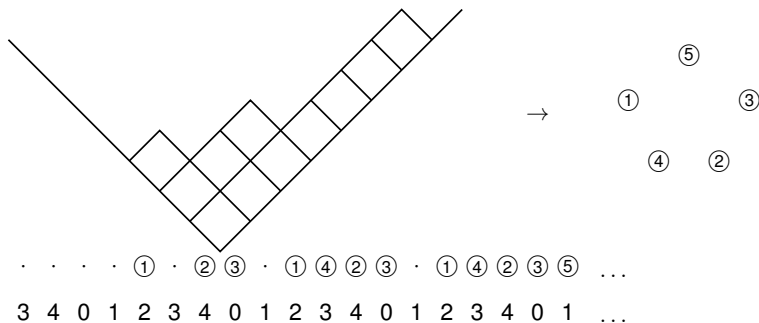
The proof of this result used a bijection to a TASEP-model with cyclic k -permutations of length k . Example with 5-cores.



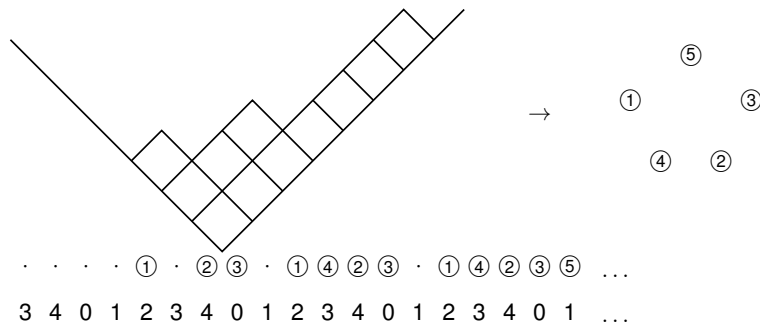
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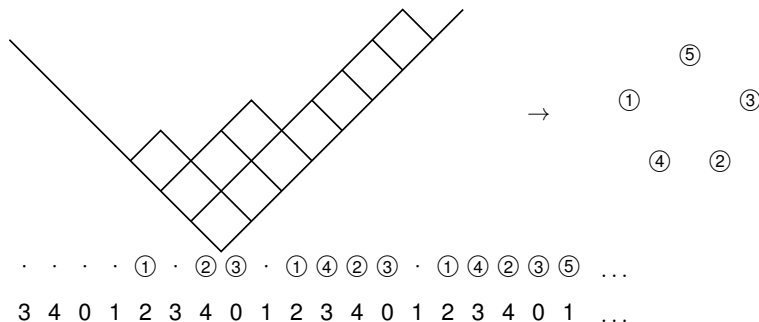


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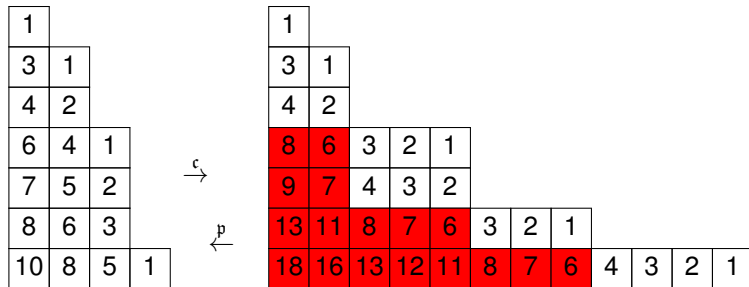
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The proof computed certain correlations in the stationary distribution of this Markov chain called TASEP. Using beautiful theory of Ferrari-Martin.

Question: Can we do the same thing for k -cores and the Plancherel measure?

k -bounded partitions

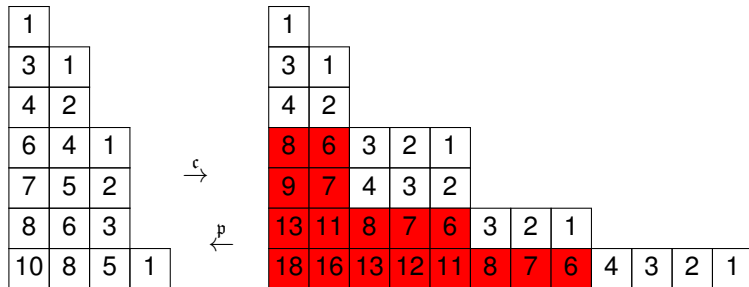
A partition $\lambda : \lambda_1 \geq \dots \geq \lambda_r > 0$ with $k \geq \lambda_1$ is called a **k -bounded partition**. There is a natural bijection between k -bounded partitions and $k + 1$ -cores with the same number of parts.



The Young diagram of a 4-bounded partition λ and the map to a 5-core partition κ with hook lengths. The red boxes have hook length greater than 5 and are removed by p and added by c .

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Fact: Adding boxes to all corners with same content $i \pmod{k+1}$ in κ corresponds to adding one box to $\lambda = p(\kappa)$.

Schur functions

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Here, first let the homogeneous symmetric functions for a set of possibly infinite variables $x = \{x_1, x_2, \dots\}$ be defined as

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Extend the definition of h_m to any partition $\lambda = (\lambda_1, \dots, \lambda_{l(\lambda)})$ of $\lambda_1 + \dots + \lambda_{l(\lambda)}$ by letting $h_\lambda(x) = h_{\lambda_1}(x) \dots h_{\lambda_{l(\lambda)}}(x)$.

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Now, we may define the Schur functions via the Jacobi-Trudi formula:

$$s_\lambda(x) = \det(h_{\lambda_i + j - i}(x))_{1 \leq i, j \leq n}.$$

Schur and k -Schur functions

A third way is as follows. Define raising operators R_{ij} by

$$R_{ij}(\lambda) = (\lambda_1, \dots, \lambda_i + 1, \dots, \lambda_j - 1, \dots, \lambda_{l(\lambda)}) \quad (1)$$

for $1 \leq i \leq j \leq l(\lambda)$, and we let $R_{ij}h_\lambda = h_{R_{ij}(\lambda)}$.

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Now, for a k -bounded partition λ , we can define the k -Schur function

$$s_\lambda^{(k)} = \prod_{i=1}^{l(\lambda)} \prod_{j=i+1}^{k-\lambda_i+i} (1 - R_{ij}) h_\lambda.$$

Example of Shur and k -Schur function

Let $\lambda = (2, 2, 1)$. For the classical Schur function $s_{(2,2,1)}$, expanding the product $(1 - R_{12})(1 - R_{13})(1 - R_{23})h_{(2,2,1)}$ gives:

$$\begin{aligned}s_{(2,2,1)} &= h_{(2,2,1)} - h_{R_{12}(\lambda)} - h_{R_{13}(\lambda)} - h_{R_{23}(\lambda)} + h_{R_{12}R_{13}(\lambda)} \\&\quad + h_{R_{12}R_{23}(\lambda)} + h_{R_{13}R_{23}(\lambda)} - h_{R_{12}R_{13}R_{23}(\lambda)} \\&= h_{(2,2,1)} - h_{(3,1,1)} - h_{(3,2,0)} - h_{(2,3,0)} + h_{(4,1,0)} \\&\quad + h_{(3,2,0)} + h_{(3,3,-1)} - h_{(4,2,-1)} \\&= h_{(2,2,1)} - h_{(3,1,1)} - h_{(2,3,0)} + h_{(4,1,0)}.\end{aligned}$$

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While for the k -Schur function with $k = 3$, the operator R_{13} is excluded because $j = 3 > k - \lambda_1 + 1 = 2$. The product simplifies to $(1 - R_{12})(1 - R_{23})h_{(2,2,1)}$, which gives:

$$\begin{aligned}s_{(2,2,1)}^{(3)} &= h_{(2,2,1)} - h_{R_{12}(\lambda)} - h_{R_{23}(\lambda)} + h_{R_{12}R_{23}(\lambda)} \\&= h_{(2,2,1)} - h_{(3,1,1)} - h_{(2,3,0)} + h_{(3,2,0)}.\end{aligned}$$

Strong dimension

The dimension f_λ is a coefficient in the Schur function s_λ ,

$$f_\lambda = [x_1 x_2 \dots x_n] s_\lambda.$$

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One may compute d_λ as certain strong marked tableaux of shape λ .

Theorem (Lam, Lapointe, Morse, Shimozono 2010)

Summing over all k -bounded partitions of n we have

$$\sum_{\lambda} f_\lambda^{(k)} d_\lambda^{(k)} = n!.$$

So this gives a probability measure on k -bounded partitions, the **k -Plancherel measure**.

k -Plancherel measure process

Now define the following growth in the weak lattice of k -bounded partitions (equivalently $k + 1$ -cores). Let

$$P(\lambda \rightarrow_k \Lambda) = \frac{d_\Lambda^{(k)}}{n d_\lambda^{(k)}},$$

we get measure $\pi(\lambda) = f_\lambda^{(k)} \cdot d_\lambda^{(k)} / n!$, since

$$\pi(\Lambda) = \sum_{\lambda \rightarrow_k \Lambda} \pi(\lambda) P(\lambda \rightarrow_k \Lambda) = \sum_{\lambda \rightarrow_k \Lambda} \frac{f_\lambda^{(k)} \cdot d_\lambda^{(k)}}{(n-1)!} \frac{d_\Lambda^{(k)}}{n d_\lambda^{(k)}} = \frac{f_\Lambda^{(k)} d_\Lambda^{(k)}}{n!}$$

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Note that $f_\lambda^{(k)} \leq f_\lambda \leq d_\lambda^{(k)}$

k -Rectangle property

Let $\square_i$ be the partition $(i, i, \dots, i)$ with $k - i + 1$ parts for $i = 1, \dots, k$.

We have :

k -rectangle property (Lapointe-Morse 2007): For any partition $\lambda \in \mathcal{P}_k$ and $i \in [k]$,

$$s_{\square_i}^{(k)} s_{\lambda}^{(k)} = s_{\lambda \cup \square_i}^{(k)}.$$

Thus also

$$d_{\square_i}^{(k)} d_{\lambda}^{(k)} = d_{\lambda \cup \square_i}^{(k)}.$$

The infinite process thus repeats. And we can remove any $\square_i$ that occurs. For $\lambda = (\lambda_1, \lambda_2, \dots)$, let $l_i(\lambda)$ denote the number of parts with length i , and set

$$\mathcal{R}_k := \{\lambda \text{ } k\text{-bounded} : l_i(\lambda) \leq k - i \text{ for all } i = 1, \dots, k\}.$$

Since $|\mathcal{R}_k| = k!$, we obtain a Markov chain on $k!$ partitions, or equivalently on the $k!$ cyclic $k + 1$ -permutations.

The transition probability for any $\lambda, \mu \in \mathcal{R}_k$ is given by

$$P(\lambda, \mu) = \frac{1}{|\lambda| + 1} \times \begin{cases} d_{\mu}^{(k)} / d_{\lambda}^{(k)} & \text{if } \lambda \rightarrow_k \mu, \\ d_{\Lambda}^{(k)} / d_{\lambda}^{(k)} & \text{if } \lambda \rightarrow_k \Lambda = \mu \cup \square_i \text{ for some } i \in [k], \\ 0 & \text{otherwise.} \end{cases}$$

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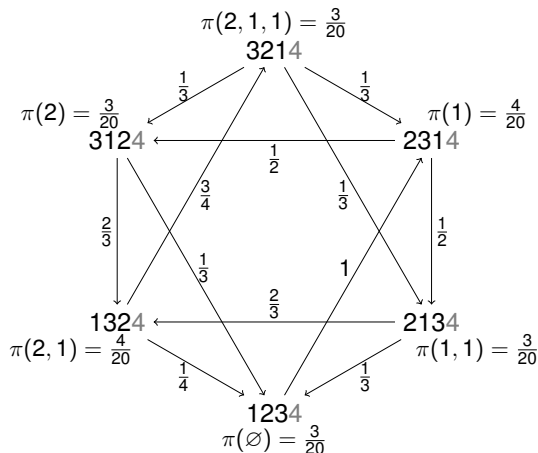
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Theorem (L.-Özdemir)

For $\lambda, \mu \in \mathcal{R}_k$, μ can be obtained from λ by adding a box to the column i of λ if and only if particle i can jump clockwise in the strong TASEP^(k). In particular, if i has a smaller number in front (clockwise) in TASEP^(k), then adding a box to column i of λ is not a valid transition in the chain.

The Markov chain TASEP⁽³⁾



A result

Let π be stationary distribution of $\text{TASEP}^{(k)}$. We want to learn as much as possible about π .

Define **k -conjugation** of $\lambda \in \mathcal{R}_k$ as $\lambda^{\omega_k} = p(c(\lambda)')$. We can prove **conjugation symmetry**.

Theorem (L-Özdemir)

For all $\lambda \in \mathcal{R}_k$, $d_{\lambda}^{(k)} = d_{\lambda^{\omega_k}}^{(k)}$ and $\pi(\lambda) = \pi(\lambda^{\omega_k})$.

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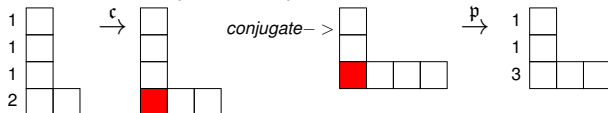
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Example $\lambda = (2, 1, 1, 1) \in \mathcal{R}_4$.



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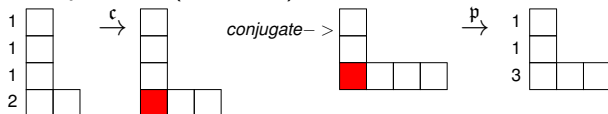
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So $\pi(2, 1, 1, 1) = \pi(3, 1, 1)$

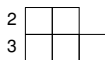
A conjecture

For a $\lambda \in \mathcal{R}_k$, let $\lambda^{\text{comp}} \in \mathcal{R}_k$ be such that $\lambda \cup \lambda^{\text{comp}} = (k-1, (k-2)^2, \dots, i^{k-i}, \dots, 2^{k-2}, 1^{k-1})$, that is $l_i(\lambda) + l_i(\lambda^{\text{comp}}) = k - i$.

Example $\lambda = (2, 1, 1, 1) \in \mathcal{R}_4$.

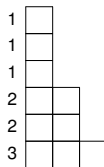


λ



λ^{comp}

Total



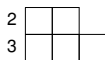
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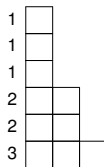


λ



λ^{comp}

Totalt



Conjecture (Symmetry of complements)

For all $\lambda \in \mathcal{R}_k$, $\pi(\lambda) = \pi(\lambda^{\text{comp}})$.

π for TASEP⁽⁴⁾

$\alpha^{-1}(\lambda)$	λ	$\pi(\lambda) \cdot 280$	$\alpha^{-1}(\lambda^{\text{comp}})$	λ^{comp}
1234	$\emptyset$	8	4321	3 2 2 1 1 1
2341	1	14	1432	3 2 2 1 1
2314	1 1	12	4132	3 2 2 1
2134	1 1 1	8	4312	3 2 2
3412	2	12	2143	3 2 1 1 1
3142	2 1	16	2413	3 2 1 1
1342	2 1 1	15	2431	3 2 1
3421	2 1 1 1	12	1243	3 2
3124	2 2	8	4213	3 1 1 1
1324	2 2 1	12	4231	3 1 1
3241	2 2 1 1	15	1423	3 1
3214	2 2 1 1 1	8	4123	3

The stationary probabilities are much nicer than the transitions rates!

More conjectures

The greatest common denominators in π are 20, 280, 70560, 310464 for $k = 3, 4, 5, 6$ respectively. Let $M_k = \prod_{j=1}^k \binom{2j}{j}$.

Conjecture

For every $\lambda \in \mathcal{R}_k$, we have $\pi(\lambda) = A_\lambda / M_k$, for some integer A_λ .

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Conjecture

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Conjecture

The minimal value of π in the k -chain is $(k+1) \prod_{j=1}^k \binom{k}{j} / M_k$. And moreover, the minimum value is obtained exactly for those partitions λ where, for each i we have either $l_i(\lambda) = 0$ or $l_i(\lambda) = i - 1$. In particular the minimum is obtained exactly 2^{k-1} times.

Let ρ_i be the expected value in $\text{TASEP}^{(k)}$ that $\square_i$ is formed and removed.

Conjecture

$$\rho_i = \frac{1}{\binom{k+2}{3}} \text{ for all } i \in [k].$$

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Conjecture

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We can prove

Theorem

Let $\lambda \in \mathcal{R}_k$ with $l_1(\lambda) = k - 1$ and $\mu = \lambda \setminus (1^{k-1})$. Then, we have $P(\lambda, \mu) = 1/k$. That is to say, adding a box in the first column of a partition with $(k - 1)$ parts with size 1, so that we remove $\square_1$, is always $1/k$.

But to prove the conjecture we would also need $\sum_{\lambda} \pi(\lambda)$, where the sum is over all λ with $l_1(\lambda) = k - 1$.

The infinite chain

Conjecture

The limit shape of the $k + 1$ -core in the $k + 1$ -Plancherel measure is D_{k+1} (the same as for uniform growth).

This last was surprising to us.

T	H	A	N	K	S														
				F	O	R													
					Y	O	U	R											
						A	T	T	E	N	T	I	O	N	!				